

115.109

Introduction to Business Analytics

Topic 8:

Determining Difference: Means

Dr Andrew Murphy

(Please contact me via Communication Tools in Stream)

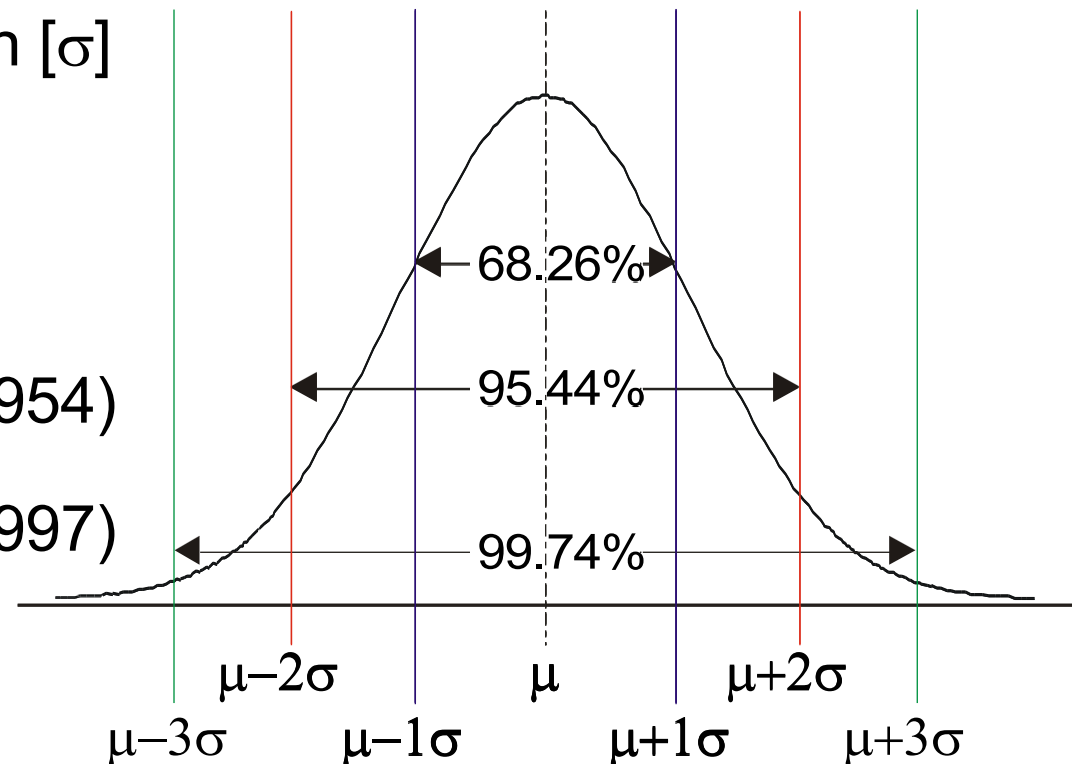
Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assessments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

Assessments Schedule

2026	Assessment	Lab 1&2	Lab 3&4	Lab 5&6	Test 1	Lab 7&8	Lab 9,11	Test 2	Report
Week	13-Jul	O: 13.07							
commencing	20-Jul		O: 20.07						
Monday:	27-Jul								
	03-Aug	D: 03.08							
	10-Aug			O: 10.08					
	17-Aug		D: 17.08		O: 14.08 C: 21.08				O 17.08
	<i>Mid-Semester Study Period</i>			D:31.08					
	07-Sep					O: 07.09			
	14-Sep								
	21-Sep						O: 21.09		
	28-Sep								
	05-Oct					D: 05.10		O: 02.10 C: 09.10	
	12-Oct						D: 19.10		D: 19.10
		O=opens (available from); D=due (without late penalty); C=closes (not available after)							

Recap: Normal Distribution

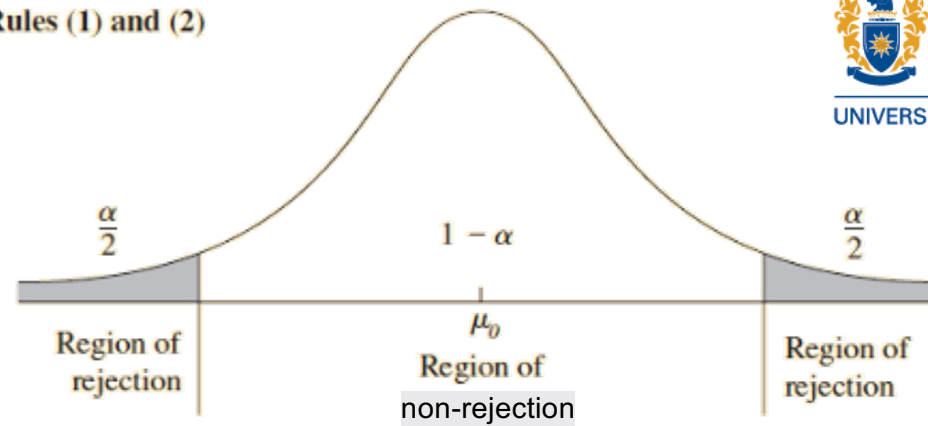
- 68.3% of observations fall within ± 1 standard deviation $[\sigma]$ of a normally distributed population mean $[\mu]$ (ie, Probability=0.683)
- 95.4% fall within $\pm 2\sigma$ ($P=0.954$)
- 99.7% fall within $\pm 3\sigma$ ($P=0.997$)



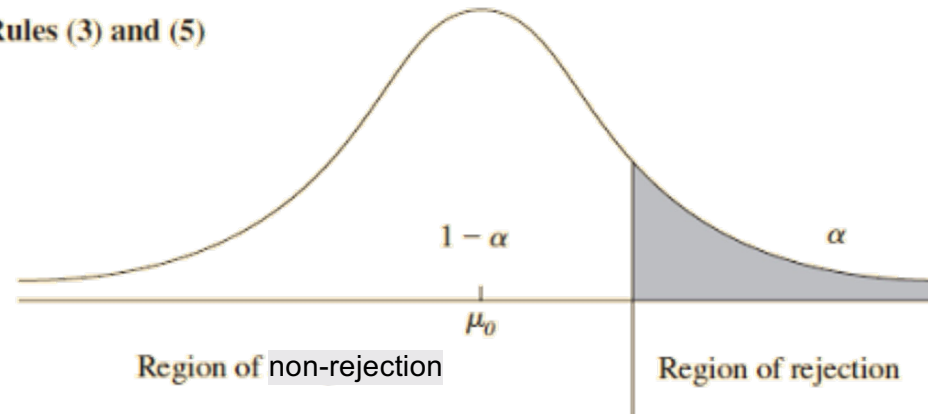
- 95% fall within $\pm 1.96\sigma$ ($P=0.95$) $-1.96\sigma \longleftrightarrow \mu \longleftrightarrow +1.96\sigma$

Outcomes

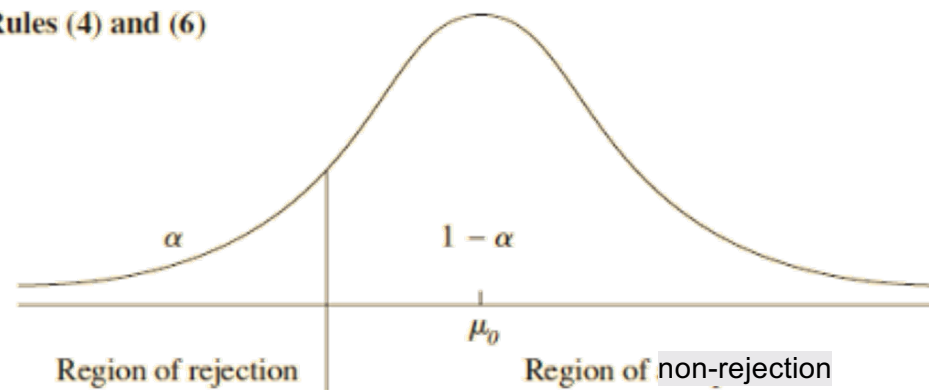
Rules (1) and (2)



Rules (3) and (5)



Rules (4) and (6)



Confidence Intervals around $\bar{X}$

Example: Bean Canning Machines: target weight = 500g

Machine samples (N=25)

$$\bar{x} = 502 \quad 498$$

$$s = 2.5 \quad 5$$

$$z = 4.0 \quad -2.0$$

$$SE_{\bar{x}} = 0.5 \quad 1.0$$

$$\underline{UCL} = 502.98 \quad 499.96$$

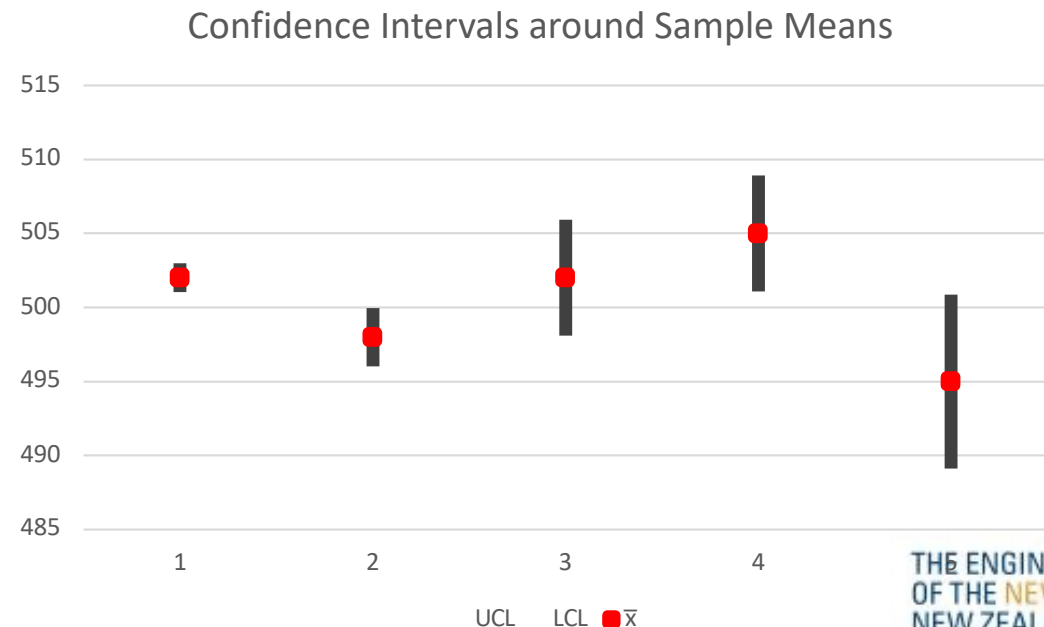
$$\underline{LCL} = 501.02 \quad 496.04$$

$$Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

$$SE_{\bar{x}} = \frac{2.5}{\sqrt{25}} = 0.5$$

$$\underline{(\bar{x} - 1.96SE_{\bar{x}})} \leq \mu \leq \underline{(\bar{x} + 1.96SE_{\bar{x}})}$$



Probability $Z \leq Z_c$
0.95

Probability $Z > Z_c$
0.05

Z_c
1.645

$Z = 2.47$

$\mu_1 = \mu_2$

Machine samples (N=25)	$\bar{x} = 502$	498
	$s = 2.5$	5

- $$H_o : \mu_1 - \mu_2 = 0$$

$$\bar{x}_1 = 502 \quad \bar{x}_2 = 498$$

- $$s_1 = 5 \quad s_2 = 10$$

$$n_1 = 40 \quad n_2 = 50$$

- $$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

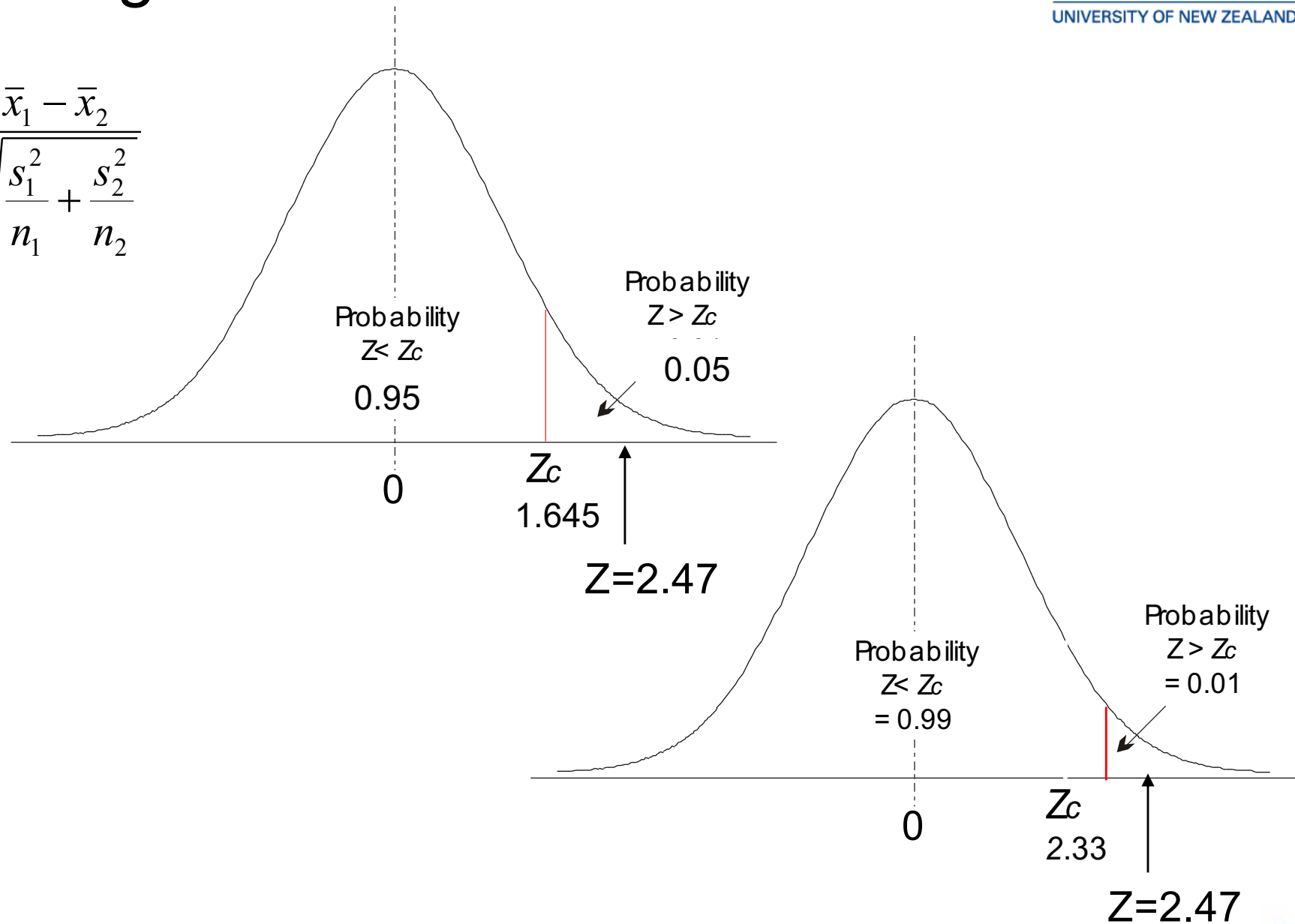
$$Z = \frac{502 - 498}{\sqrt{\frac{5^2}{40} + \frac{10^2}{50}}} = \frac{502 - 498}{\sqrt{\frac{25}{40} + \frac{100}{50}}} = \frac{4}{\sqrt{2.625}} = \frac{4}{1.62} = 2.47$$

- $$\alpha=0.05 \text{ (5\% in right tail)} \therefore z_c = 1.645$$

- THE ENGINE
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NEW ZEALAND

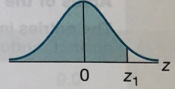
Showing Z: One-tailed multivariate

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

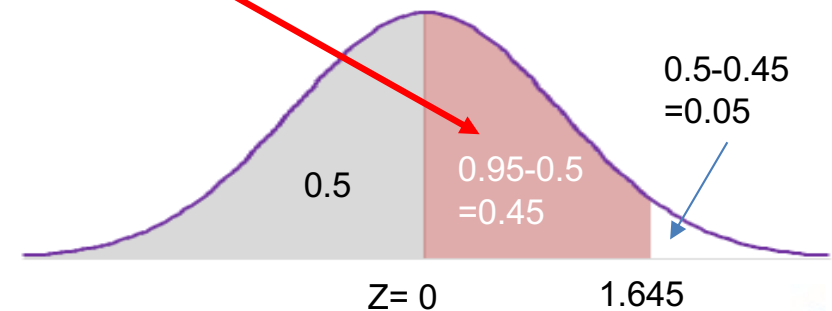
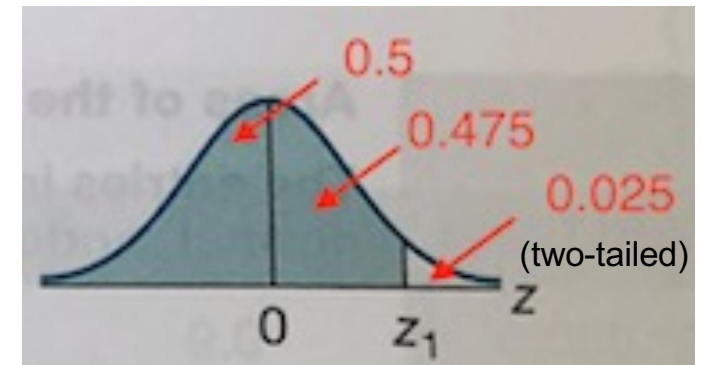


Right-tail Critical Value

TABLE A.5 Cumulative normal probabilities
The entries in this table are the cumulative probabilities that a standard normal random variable is between $-\infty$ and z_1 (the shaded area).



z_1	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952



Multivariate Hypothesis Testing

Quick method: Z-test (assumes samples large enough to be considered populations)

Best method: t-tests (samples can be small or large)

Eg: Supplier Domain: are there gender differences?

(Lab 7 Task 2,
except one-tailed)

Is supplier sustainability more important for ♂ than ♀?

1. Null Hypothesis: two samples are the same $H_0: \mu_1 = \mu_2$

$$H_0: \mu_1 - \mu_2 = 0$$

- Alternative Hypothesis: are different (two-tail) $\bar{x}_1 = 3.93$ $\bar{x}_2 = 4.08$
 $H_A: \mu_1 \neq \mu_2$ $H_A: \mu_1 - \mu_2 \neq 0$ $s_1 = 0.753$ $s_2 = 0.715$
 $n_1 = 449$ $n_2 = 591$
- or one is larger than the other (one-tail)

$$H_A: \mu_1 > \mu_2 \quad H_A: \mu_1 - \mu_2 > 0$$

31				Males			Females			
	Column	Composite Variable	Variable	Variable	Variable	Variable	Variable	Variable		
32			Mean	Standard Deviation	Sample Size	Mean	Standard Deviation	Sample Size		
33			CO	Societal Domain	3.441	0.780	449	3.56	0.703	591
34			CP	Environmental Domain	4.05	0.770	449	4.24	0.706	591
35			CQ	Supplier Domain	3.93	0.753	449	4.08	0.715	591

Multivariate Hypothesis Testing

Quick method: Z-test (assumes samples large enough to be considered populations)

Best method: t-tests (samples can be small or large)

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 $n_1 = 449$ $n_2 = 591$
- or one is larger than the other (one-tail)

$$H_A: \mu_1 > \mu_2 \quad H_A: \mu_1 - \mu_2 > 0$$

$$2. \text{ Test Statistic: } Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad Z = \frac{3.93 - 4.08}{\sqrt{\frac{0.753^2}{449} + \frac{0.715^2}{591}}} = \frac{-0.16}{\sqrt{\frac{0.567}{449} + \frac{0.511}{591}}}$$

$$= \frac{-0.16}{\sqrt{0.0013 + 0.0009}} = \frac{-0.16}{0.046} = -3.45$$

3. Compare test statistic vs critical level of significance
 $\alpha = 0.05$ (5% in right tail) $\therefore z_c = 1.645$

Multivariate Hypothesis Testing

Quick method: Z-test (assumes samples large enough to be considered populations)

Best method: t-tests (samples can be small or large)

Eg: Supplier Domain: are there gender differences?

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$$s_1 = 0.753 \quad s_2 = 0.715$$

$$n_1 = 449 \quad n_2 = 591$$

• or one is larger than the other (one-tail)

$$H_A: \mu_1 > \mu_2 \quad H_A: \mu_1 - \mu_2 > 0$$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{-0.16}{0.046} = -3.45$$

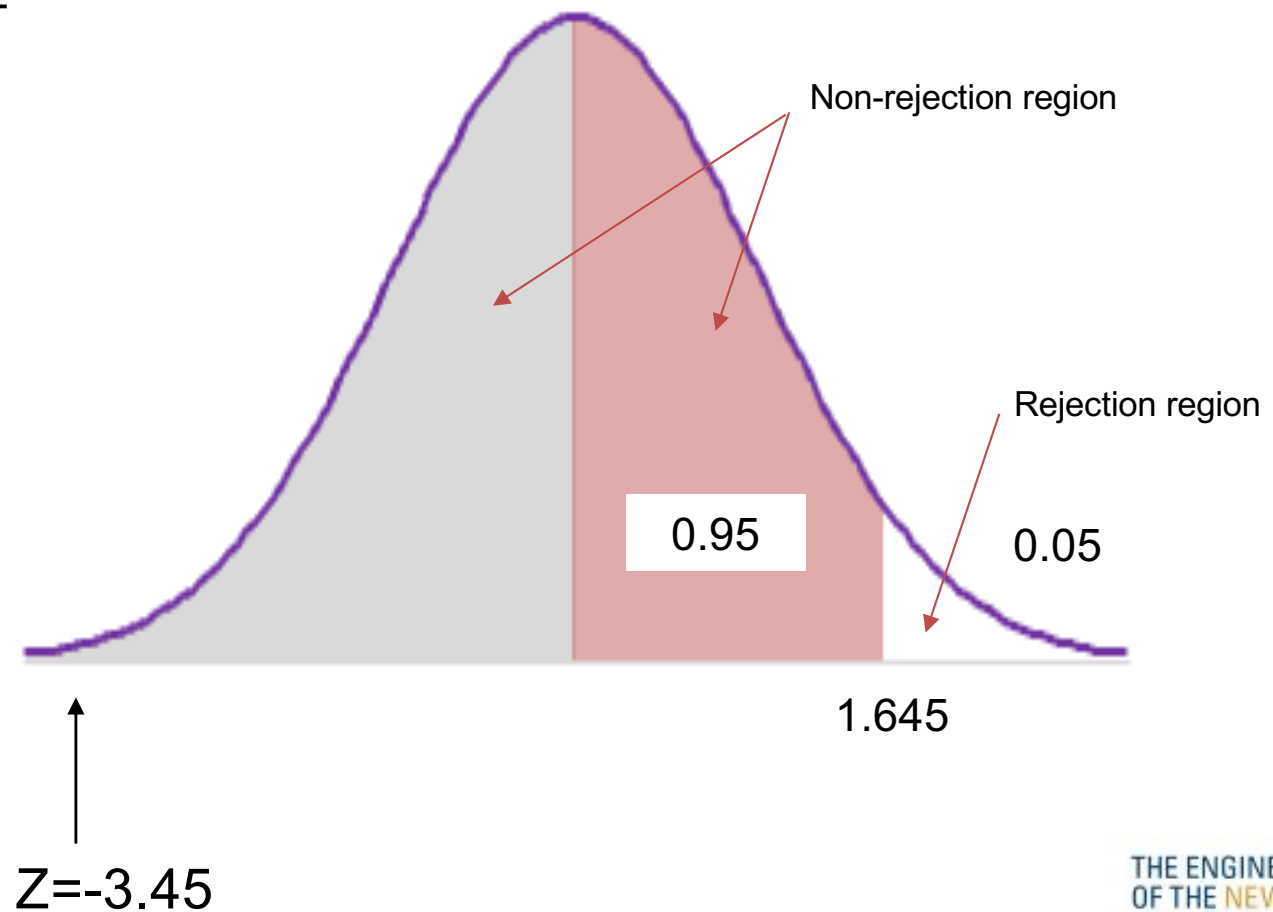
31			Males			Females					
32	Column	Composite Variable	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	Numerator	Denominator	Z-test
33	CO	Societal Domain	3.441	0.780	449	3.56	0.703	591	-0.12	0.047	-2.633
34	CP	Environmental Domain	4.05	0.770	449	4.24	0.706	591	-0.1916743	0.04650381	-4.122
35	CQ	Supplier Domain	3.93	0.753	449	4.08	0.715	591	-0.1588414	0.046104621	-3.445
36	CR	Shareholder Domain	4.02	0.716	449	4.09	0.700	591			
37	CS	Community Domain	3.81	0.721	449	3.97	0.703	591			
38	CT	Employee Domain	4.22		449	4.38	0.72	591			
39	CU	Customer Domain	3.87	0.769	449	4.12	0.681	591			
40	CV	Sustainability literacy	3.41	0.749	449	3.51	0.676	591			

4. Decision: cannot reject H_0 as not in rejection region

Showing Z: One-tailed

$$H_A: \mu_1 > \mu_2 \quad H_A: \mu_1 - \mu_2 > 0$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$



Difference in Means: Excel

Add in Analysis Toolpak: Mac: Tools > Excel Add-ins > Analysis Toolpak
PC: File > Options > Add-Ins > Manage Excel Add-Ins > Go... > Analysis Toolpak

**Lab 7 Task 2, but using the Toolpak
(rather than Subtotals with filters)**

z-Test: Two Sample for Means

Input

Variable 1 Range:

Supplier Domain by Gender

\$CQ\$5:\$CQ\$453

Variable 2 Range:

\$CQ\$454:\$CQ\$1044

Hypotheses

0

Variable 1

= 0.567

Column	Composite Variable	Variable Mean	Males Variable Standard Deviation
CO	Societal Domain	3.44	0.780
CP	Environmental Domain	4.05	0.770
CQ	Supplier Domain	3.93	0.753

Variable 2 Variance (known):

0.715

x

0.715

= 0.511

☐ Labels

Alpha:

0.05

z-Test: Two Sample for Means

	Variable 1	Variable 2
Mean	3.92576095	4.08460237
Known Variance	0.567	0.511
Observations	449	591
Hypothesized	0	
z	-3.4437756	
P(Z<=z) one-tail	0.00028683	(if left-tail)
z Critical one-tail	1.64485363	
P(Z<=z) two-tail	0.00057365	
z Critical two-tail	1.95996398	

Difference in Means: Excel

Add in Analysis Toolpak: Mac: Tools > Excel Add-ins > Analysis Toolpak
PC: File > Options > Add-Ins > Manage Excel Add-Ins > Go... > Analysis Toolpak

**Lab 7 Task 2, but using the Toolpak
(rather than Subtotals with filters)**

z-Test: Two Sample for Means

Input

Variable 1 Range:

Variable 2 Range:

Hypothesized Mean Difference:

Variable 1 Variance (known):

Variable 2 Variance (known):

☐ Labels

Alpha:

Supplier Domain by Gender

\$CQ\$5:\$CQ\$453

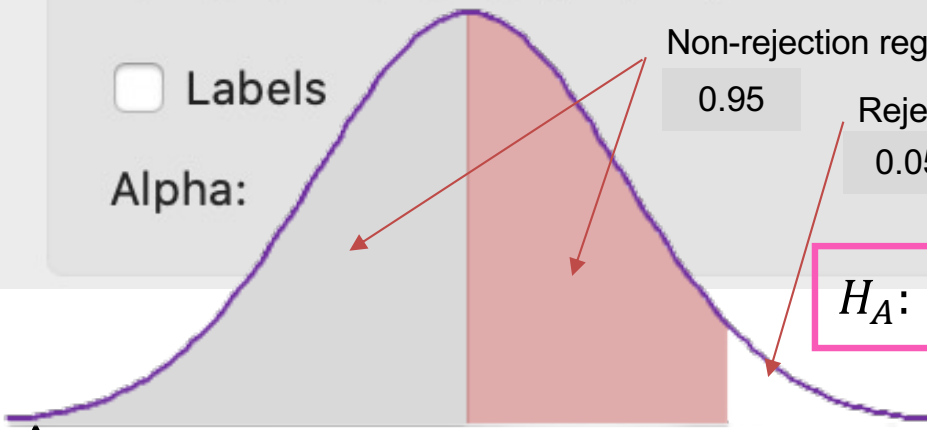
\$CQ\$454:\$CQ\$1044

0

0.567

0.511

0.05



Non-rejection region

0.95

Rejection region

0.05

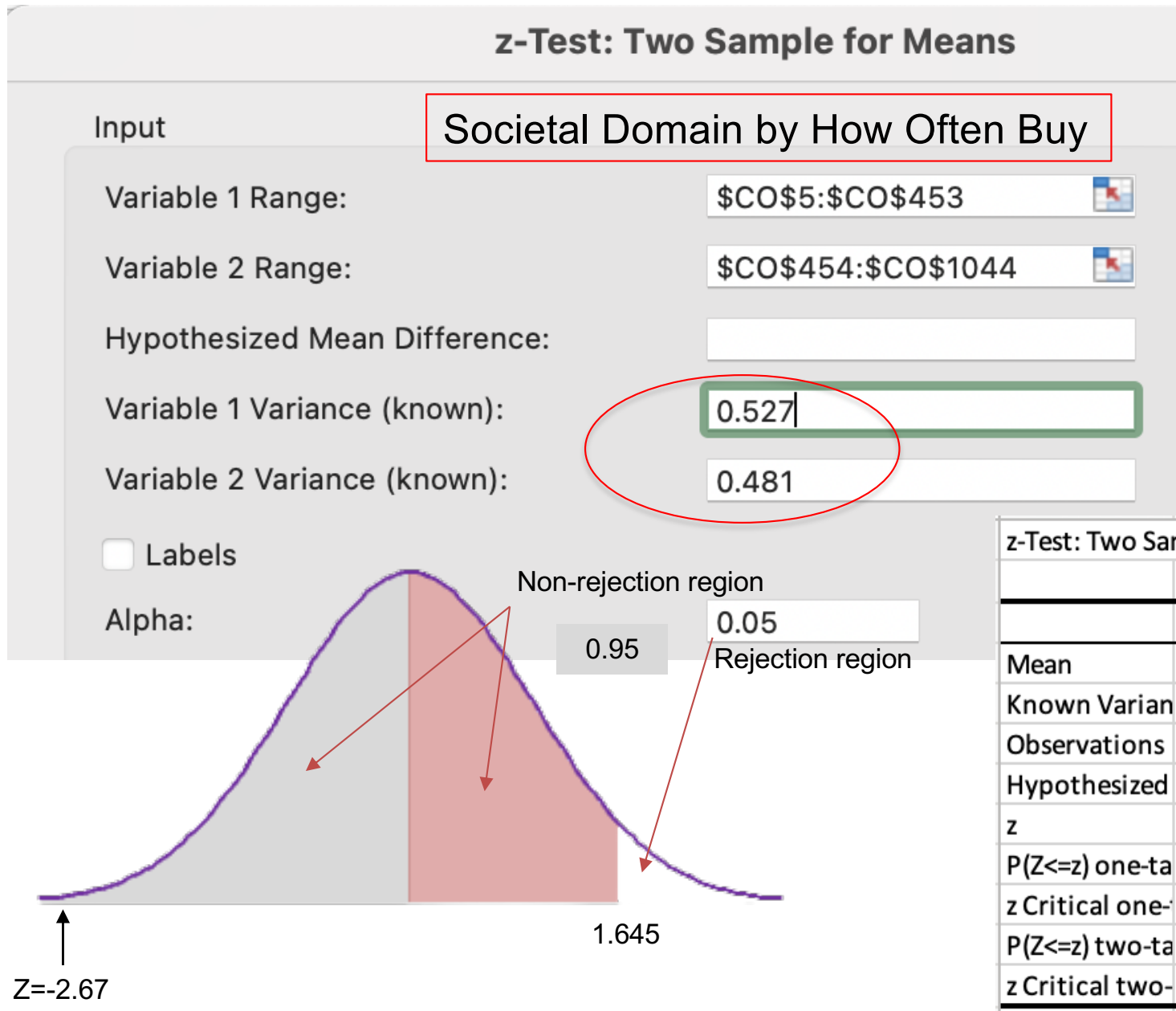
$H_A: \mu_1 > \mu_2$

	Variable 1	Variable 2
Mean	3.92576095	4.08460237
Known Variance	0.567	0.511
Observations	449	591
Hypothesized Mean Difference	0	
z	-3.4437756	
P(Z<=z) one-tail	0.00028683	(if left-tail)
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z Critical two-tail	1.95996398	

Difference in Means: Excel



Like Lab 7 Task 4, using the Toolpak (but for a different domain)

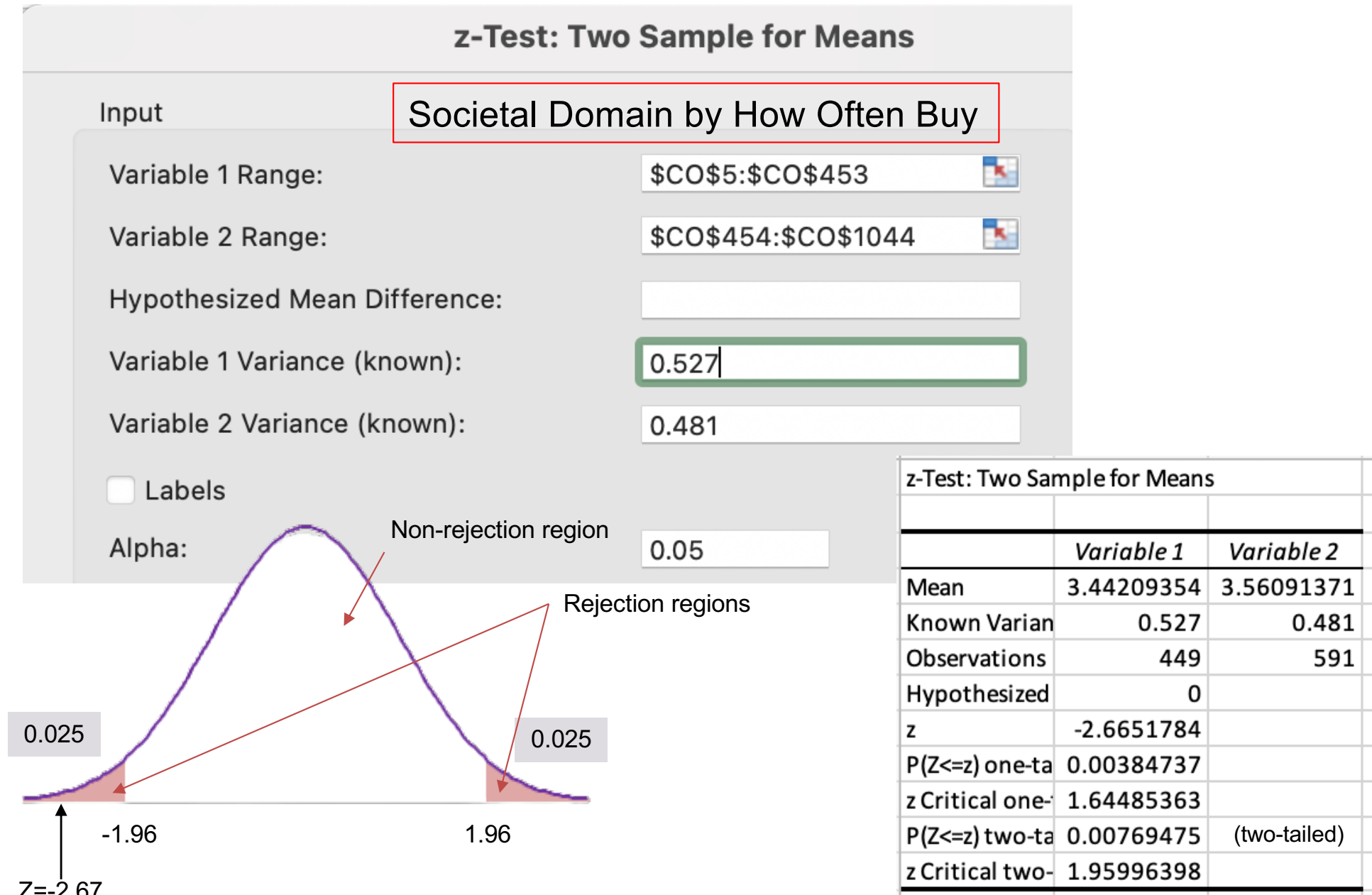


z-Test: Two Sample for Means		
	Variable 1	Variable 2
Mean	3.44209354	3.56091371
Known Variance	0.527	0.481
Observations	449	591
Hypothesized Mean Difference	0	
z	-2.6651784	
P(Z<=z) one-tail	0.00384737	(if left-tail)
z Critical one-tail	1.64485363	
P(Z<=z) two-tail	0.00769475	
z Critical two-tail	1.95996398	

Difference in Means: Excel



Like Lab 7 Task 4, using the Toolpak (but for a different domain)

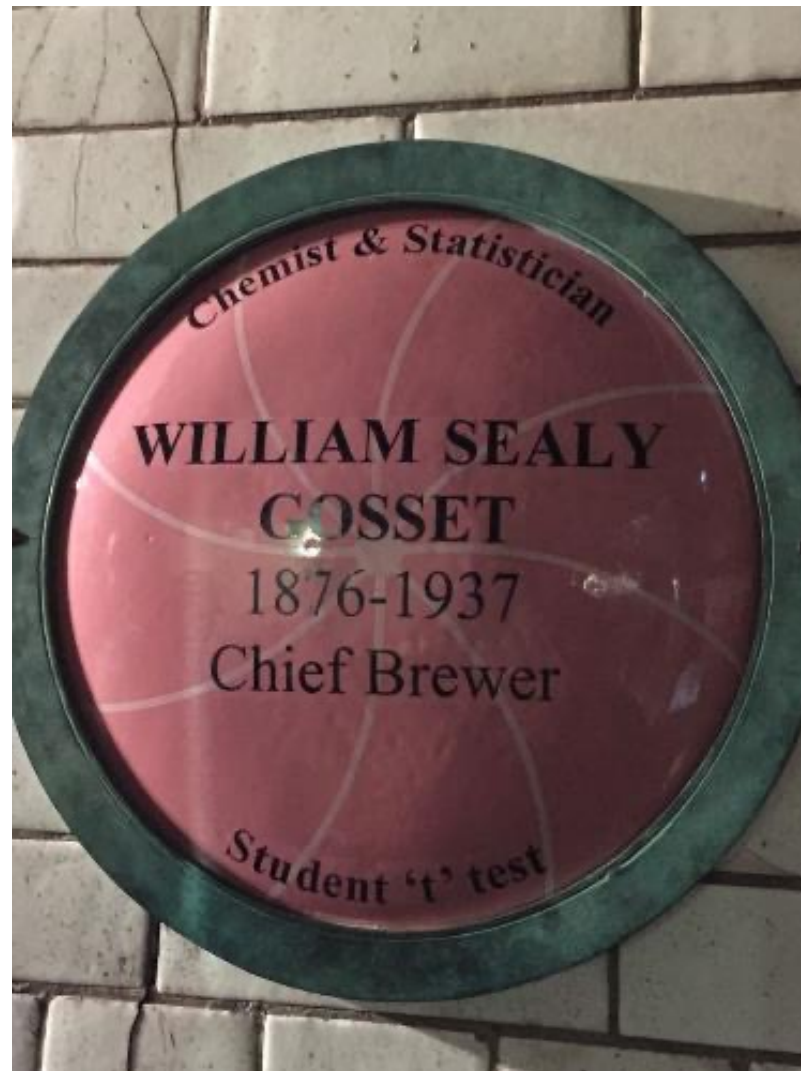


The t -distribution

- Used in most statistical tests / packages (eg SPSS)
- The t -distribution is a symmetrical, bell-shaped distribution that is contingent on sample size.
 - It has a mean of zero and a standard deviation equal to one (just like Z)
- The shape of the t -distribution is influenced by its degrees of freedom.
 - Number of observations minus number of constraints or assumptions
 - For sample sizes over 30, t -distribution begins to closely approximate the Z -distribution.

Student's T-distribution

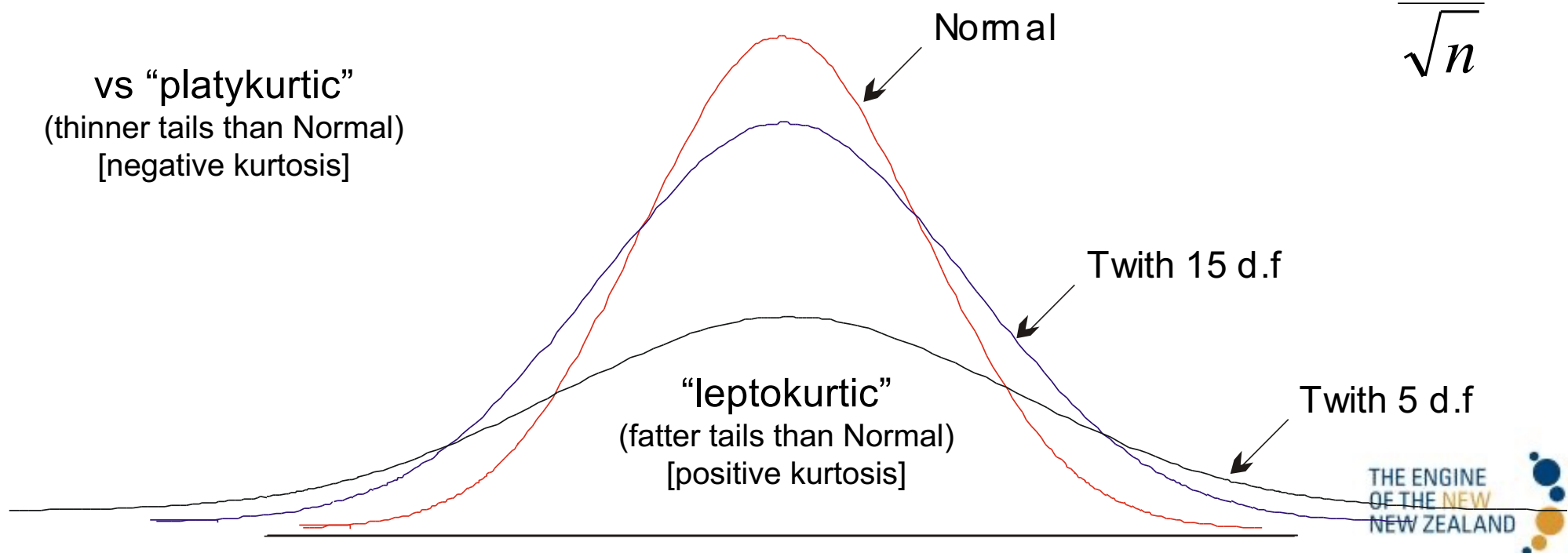
- A Student: pen name of William Gossett (Guinness brewery statistician)



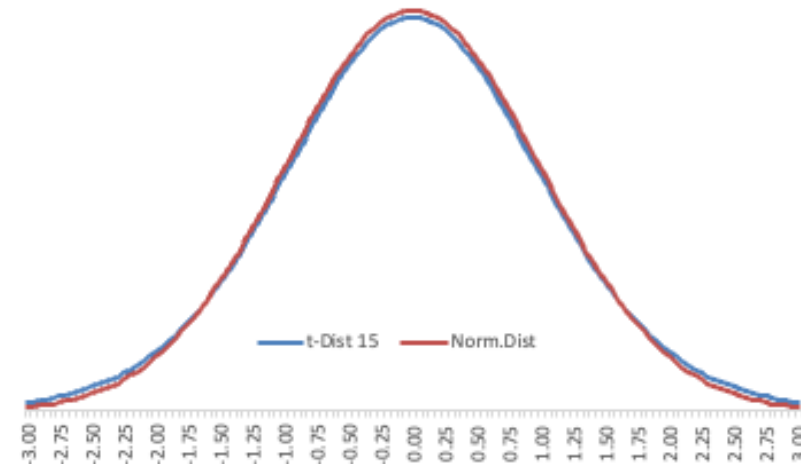
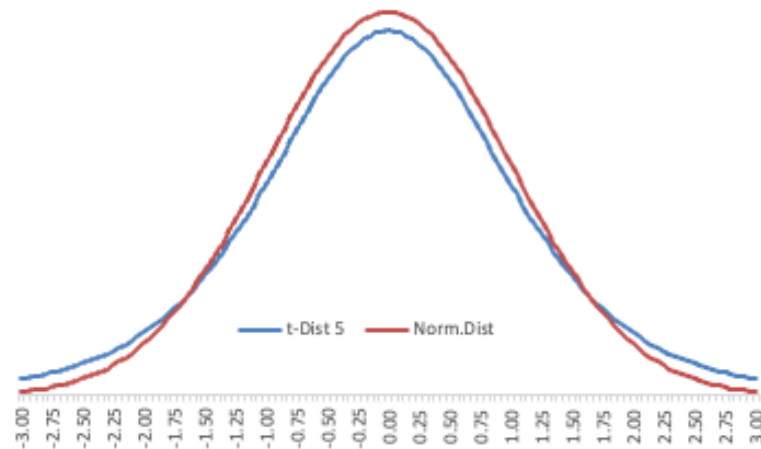
Student's T-distribution: univariate

- Ideal when n is “small” (eg < 30); and σ unknown
- In practice: used in most statistical tests
- Same test statistic formula (but termed “t”):
- Also “bell shaped”, symmetrical – but flatter than Normal
- Shape determined by “degrees of freedom” ($n-1$)

$$t_{df=n-1} = \frac{x - \bar{x}}{\frac{s}{\sqrt{n}}}$$



Student's T-distribution



For $\alpha=0.05$ (95%)

Degrees of freedom	t-value		Degrees of freedom	t-value
1	12.71		25	2.06
2	4.30		30	2.04
5	2.57		40	2.02
10	2.23		60	2.00
15	2.13		120	1.98
20	2.09		∞	1.96

Excel function:

T.INV(probability,deg_freedom)

T.INV(0.05,30) = 2.042

T.INV(0.05,100) = 1.984

See also T.TEST; T.DIST

TDIST(x,deg_freedom,tails)

TDIST(2,100,2)=0.048

Goal: <0.05

t-Table (Croucher p. 785)

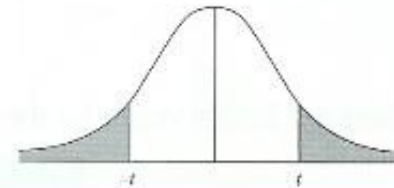


Table 7 Critical values for the t-distribution

α	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001	0.0005	α
ν										ν
1	3.078	6.314	12.71	31.82	63.66	127.3	318.3	636.6	1273	1
2	1.886	2.920	4.303	6.965	9.925	14.09	22.33	31.60	44.70	2
3	1.638	2.353	3.182	4.541	5.841	7.453	10.22	12.92	16.33	3
4	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610	10.31	4
5	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869	7.946	5
6	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959	6.798	6
7	1.415	1.895	2.355	2.998	3.499	4.029	4.785	5.408	6.082	7
8	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041	5.617	8
9	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.761	5.291	9
10	1.372	1.812	2.228	2.754	3.169	3.581	4.144	4.587	5.049	10
11	1.363	1.796	2.201	2.718	3.105	3.497	4.025	4.437	4.863	11
12	1.356	1.782	2.179	2.681	3.055	3.420	3.930	4.318	4.716	12
13	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221	4.597	13
14	1.345	1.761	2.145	2.624	2.977	3.325	3.787	4.140	4.499	14
15	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073	4.427	15
16	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015	4.346	16
17	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.955	4.286	17
18	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922	4.233	18
19	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.893	4.187	19
20	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.860	4.146	20
21	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819	4.110	21
22	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792	4.077	22
23	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.768	4.047	23
24	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745	4.021	24
25	1.316	1.708	2.060	2.485	2.787	3.079	3.450	3.725	3.996	25
26	1.315	1.706	2.055	2.479	2.779	3.067	3.435	3.707	3.974	26
27	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690	3.954	27
28	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674	3.935	28
29	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659	3.918	29
30	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646	3.902	30
35	1.306	1.690	2.030	2.438	2.724	2.996	3.340	3.591	3.836	35
40	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551	3.788	40
45	1.301	1.679	2.014	2.412	2.690	2.952	3.281	3.520	3.752	45
50	1.299	1.676	2.009	2.403	2.678	2.937	3.261	3.496	3.723	50
60	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460	3.681	60
70	1.294	1.667	1.994	2.381	2.648	2.899	3.211	3.435	3.651	70
80	1.292	1.664	1.990	2.374	2.639	2.887	3.195	3.416	3.629	80
90	1.291	1.662	1.987	2.368	2.632	2.878	3.183	3.402	3.612	90
∞	1.282	1.645	1.960	2.326	2.575	2.807	3.090	3.291	3.401	∞

Multivariate Hypothesis Testing

Quick method: Z-test (assumes samples large enough to be considered populations)

Best method: t-tests: samples can be small or large

Eg: Supplier Domain: are there gender differences?

Is supplier sustainability more important for ♂ than ♀?

1. Null Hypothesis: two samples are the same $H_0: \mu_1 = \mu_2$

$$H_0: \mu_1 - \mu_2 = 0$$

- Alternative Hypothesis: (one-tail) $H_A: \mu_1 > \mu_2$ $H_A: \mu_1 - \mu_2 > 0$

$$\bar{x}_1 = 3.93 \quad \bar{x}_2 = 4.08$$

$$s_1 = 0.753 \quad s_2 = 0.715$$

$$n_1 = 449 \quad n_2 = 591$$

2. Test Statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

3. Compare test statistic vs critical level of significance

Multivariate: Two-tailed

$$\begin{aligned}\bar{x}_1 &= 3.93 & \bar{x}_2 &= 4.08 \\ s_1 &= 0.753 & s_2 &= 0.715 \\ n_1 &= 449 & n_2 &= 591\end{aligned}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

1. $H_0: \mu_1 = \mu_2 ; H_A: \mu_1 > \mu_2$

2. Test Statistic:

$$s_p = \sqrt{\frac{(449 - 1)0.753^2 + (591 - 1)0.715^2}{449 + 591 - 2}} = 0.732$$

$$\begin{aligned}t &= \frac{3.93 - 4.08}{0.732 \sqrt{\frac{1}{449} + \frac{1}{591}}} \\ &= \frac{-0.15}{0.046} \\ t &= -3.27 \\ \{z &= -3.45\}\end{aligned}$$

3. Compare test statistic vs critical level of significance

$\alpha=0.05$ (5% in right tail)

Degrees of freedom: $n_1 + n_2 - 2 = 1038$

$\therefore t_c = 1.646$ (for d.f = 1038) $\{\simeq \text{tinv}(0.10, 1038)\}$

4. Decision: if test statistic is beyond t_c then reject H_0 :

$t < t_c : -3.27 < 1.646 \therefore$ cannot reject H_0

We cannot say the male mean is greater than the female mean for Supplier domain, with $\alpha=0.05$

t-Table (Croucher p. 785)

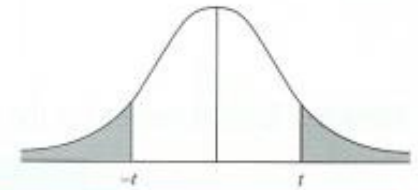
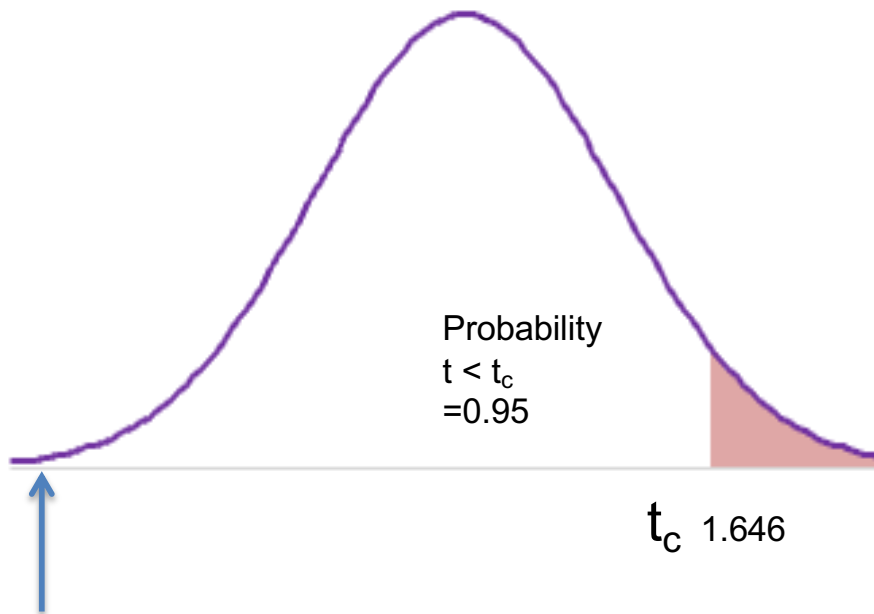


Table 7 Critical values for the t-distribution

α ↓ t	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001	0.0005	α ↑ t
1	3.078	6.314	12.71	31.82	63.66	127.3	318.3	636.6	1273	1
2	1.886	2.920	4.303	6.965	9.925	14.09	22.33	31.60	44.70	2
3	1.638	2.353	3.182	4.541	5.841	7.453	10.22	12.92	16.33	3
4	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610	10.31	4
5	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869	7.976	5
6	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959	6.788	6
7	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408	6.082	7
8	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041	5.617	8
9	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781	5.291	9
10	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587	5.049	10
11	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437	4.863	11
12	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318	4.716	12
13	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221	4.597	13
14	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140	4.499	14
15	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073	4.417	15
16	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015	4.346	16
17	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965	4.286	17
18	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922	4.233	18
19	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883	4.187	19
20	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850	4.146	20
21	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819	4.110	21
22	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792	4.077	22
23	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.768	4.047	23
24	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745	4.021	24
25	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725	3.996	25
26	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707	3.974	26
27	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690	3.954	27
28	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674	3.935	28
29	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659	3.918	29
30	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646	3.902	30
35	1.306	1.690	2.030	2.438	2.724	2.996	3.340	3.591	3.836	35
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45	1.301	1.679	2.014	2.412	2.690	2.952	3.281	3.520	3.752	45
50	1.299	1.676	2.009	2.403	2.678	2.937	3.261	3.496	3.723	50
60	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460	3.681	60
70	1.294	1.667	1.994	2.381	2.648	2.899	3.211	3.435	3.651	70
80	1.292	1.664	1.990	2.374	2.639	2.887	3.195	3.416	3.629	80
90	1.291	1.662	1.987	2.369	2.633	2.882	3.189	3.410	3.623	90



$t = -3.27$

Degrees of freedom: $n_1 + n_2 - 2 = 1038$

$\therefore t_c = 1.646$

$= \text{tinv}(0.10, 1038)$

{based on 2 tails, so double each tail to get the equivalent 1-tail result}

Difference in Means: Excel

CN	CO	CP	CQ	CR	CS	CT	CU	CV
	Societal Do	Environme	Supplier Do	Shareholde	Communit	Employee	Customer	Sustainab
	Societal	ental	Supplier	Sharehold	Communi	Employee	Customer	
Gender	Domain	Domain	Domain	er Dom	ty Dom	Domain	Domain	Sustain
Which gender do you identify with?								
1: Male	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Strongly
2: Female	important	important	important	important	important	important	important	disagree
3: Gender	5: Extremely	5: Extremely	5: Extremely	5: Extremely	5: Extremely	5: Extremely	5: Extremely	3: Neither
		Import	Import	Import	Import	Import	Import	disagree nor
								agree
								5: Strongly
								agree

t-Test: Two-Sample Assuming Equal Variances

Input

Variable 1 Range:

Variable 2 Range:

Hypothesized Mean Difference:

☐ Labels

Alpha:

t-Test: Two-Sample Assuming Equal Variance		
	Variable 1	Variable 2
Mean	3.92576095	4.08460237
Variance	0.56640086	0.51072108
Observations	449	591
Pooled Variance	0.53475243	
Hypothesized	0	
df	1038	
t Stat	-3.4696656	
P(T<=t) one-ta	0.00027126	
t Critical one-	1.64632293	
P(T<=t) two-ta	0.00054252	
t Critical two-	1.96225203	

$$t = \frac{3.93 - 4.08}{0.732 \sqrt{\frac{1}{449} + \frac{1}{591}}} = \frac{-0.15}{0.046}$$

$$t = -3.27 \text{ (rounding)}$$

$t < t_c: -3.27 < 1.646 \therefore \text{cannot reject } H_0$
 $t < t_c: -3.47 < 1.646 \therefore \text{cannot reject } H_0$

More than two subsamples: ANOVA

- **Analysis of variance (ANOVA):** Determines whether statistically significant differences in means occur between three or more groups
 - For example, whether shopping enjoyment differs for ‘young’, ‘middle-aged’ and ‘old’ respondents;
 - or ‘poor’, ‘middle’ and ‘rich’ households
- Uses variances to allow us to compare means
- If there are three groups of independent variables, the null hypothesis is as follows:

$$\mu_1 = \mu_2 = \mu_3$$

- To falsify the null hypothesis: the variation in responses *between* each of the three groups will be comparatively larger than the variation in responses *within* each of the groups.

More than two subsamples: ANOVA

- The **F-test** is used to determine whether there is more variability in the scores of one (sub)sample mean than in the scores of others. (It tests the ratio of variance **between** groups to the variance **within** groups.)
- Null hypotheses: the means being tested are not different
- Alternative hypothesis: the means are different (**at least one** is different)
- eg: Environmental domain by HowOftenBuy (1-5)

	Mean	StDev	Count	Std Error
Column (Group) 1	3.73	0.938	33	0.16
Column (Group) 2	4.11	0.642	53	0.09
Column (Group) 3	4.05	0.682	240	0.04
Column (Group) 4	4.11	0.617	261	0.04
Column (Group) 5	4.27	0.576	154	0.05

Explanation:	Sum of Squares	Degrees of freedom	Mean square	F statistic	Significance	Critical value
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	9.335	4	2.334	5.530	0.00022	2.384
Within Groups	310.609	736	0.422			
Total	319.943	740				

More than two subsamples: ANOVA

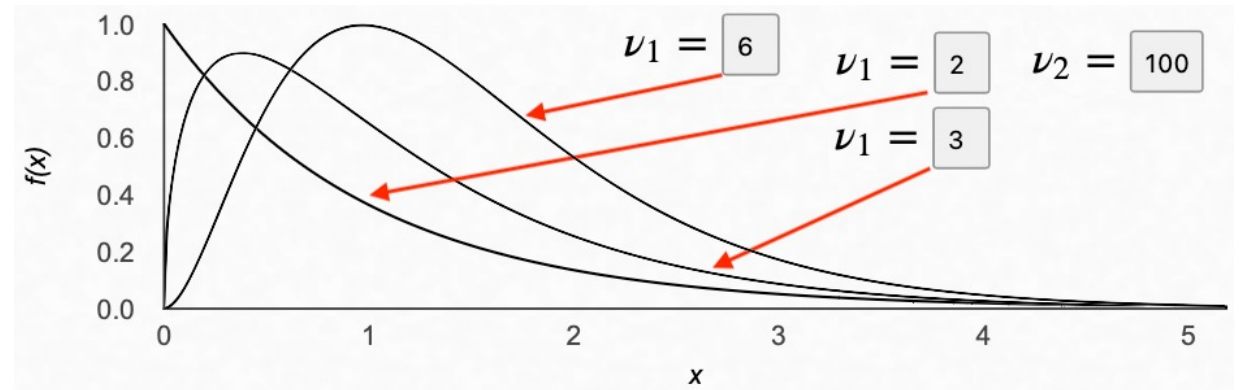
- Null hypotheses: the means being tested are not different
- Alternative hypothesis: (at least one or more) are different
- eg: Environmental Domain by HowOftenBuy

	Mean	StDev	Count	Std Error	LCL	UCL
How Often Buy =1 (Not at all)	3.73	0.938	33	0.16	3.416	4.044
How Often Buy =2	4.11	0.642	53	0.09	3.934	4.286
How Often Buy =3	4.05	0.682	240	0.04	3.972	4.128
How Often Buy =4	4.11	0.617	261	0.04	4.032	4.188
How Often Buy =5 (Very often)	4.27	0.576	154	0.05	4.172	4.368

Explanation:	Sum of Squares	Degrees of freedom	Mean square	F statistic	Significance	Critical value
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	9.335	4	2.334	5.530	0.00022	2.384
Within Groups	310.609	736	0.422			=F.INV.RT(0.05,4,736)
					=F.DIST.RT(5.53,4,736)	
Total	319.943	740				

- A calculated F-ratio (statistic) that exceeds the **critical value** indicates that the means are statistically different, and the null hypothesis is rejected
- If significance (P) value is less than 0.05, then AT LEAST ONE mean is ignificantly different. **BUT WE DON'T KNOW WHICH!**

F distribution



Explanation:	Degrees of freedom
Source of Variation	df
Between Groups	4
Within Groups	736
Total	740

- df_1 : # groups - 1 (=5-1)
- df_2 : # observations - # groups-1 (=740-4)

• $F_c \approx 2.384$

• $F = 5.53$

$\therefore$ significant:

there are significant differences between the five means

P-value: tiny

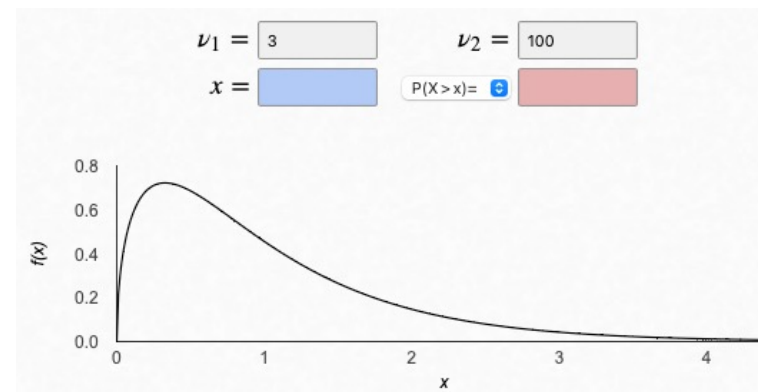
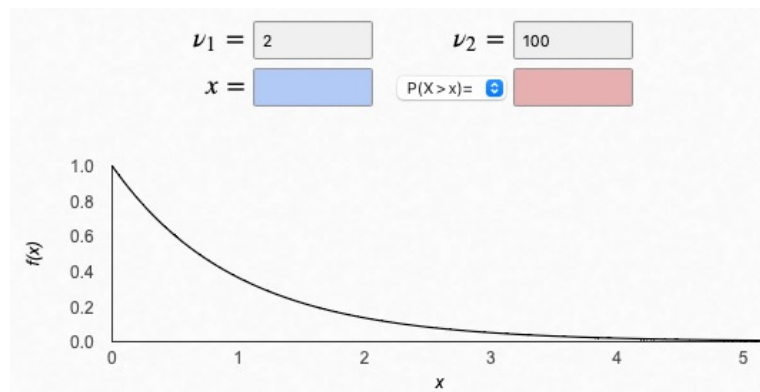
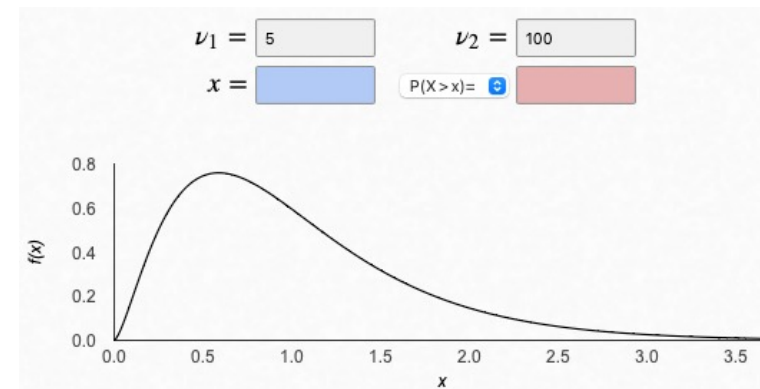
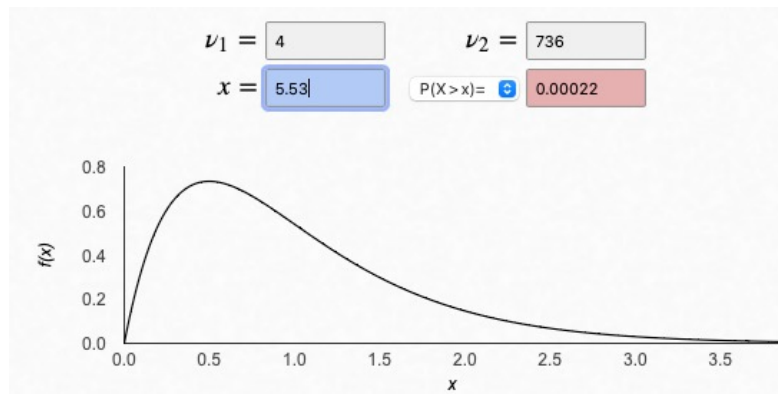
Critical value	F statistic
F_{crit}	F
2.384	5.530

Significance
P-value
0.00022

df_2	1	2	3	4	5	6	8	12	24	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	238.9	243.9	249.0	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.41	19.45	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.84	8.74	8.64	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.91	5.77	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.68	4.53	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.00	3.84	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.73	3.57	3.41	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.28	3.12	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.23	3.07	2.90	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.91	2.74	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	2.95	2.79	2.61	2.40
12	4.75	3.88	3.49	3.26	3.11	3.00	2.85	2.69	2.50	2.30
13	4.67	3.80	3.41	3.18	3.02	2.92	2.77	2.60	2.42	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.70	2.53	2.35	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.48	2.29	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.59	2.42	2.24	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.55	2.38	2.19	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.51	2.34	2.15	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.48	2.31	2.11	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.28	2.08	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.42	2.25	2.05	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.40	2.23	2.03	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.38	2.20	2.00	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.36	2.18	1.98	1.73
25	4.24	3.38	2.99	2.76	2.60	2.49	2.34	2.16	1.96	1.71
26	4.22	3.37	2.98	2.74	2.59	2.47	2.32	2.15	1.95	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.30	2.13	1.93	1.67
28	4.20	3.34	2.95	2.71	2.56	2.44	2.29	2.12	1.91	1.65
29	4.18	3.33	2.93	2.70	2.54	2.43	2.28	2.10	1.90	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.09	1.89	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.00	1.79	1.51
60	4.00	3.15	2.76	2.52	2.37	2.25	2.10	1.92	1.70	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.02	1.83	1.61	1.25
∞	3.84	2.99	2.60	2.37	2.21	2.09	1.94	1.75	1.52	1.00

F distribution shapes

<https://homepage.divms.uiowa.edu/~mbognar/applets/f.html>



ANOVA in Excel: separate into Subsamples

Sort and duplicate variable (Copy & Paste) into separate data: eg: 'Environmental Domain' (column CP), into columns DD:DH: 5x

Group 1: HowOftenBuy=1: copy CP5:CP37 **only** into DD5:DD37

Group 2: HOB=2: CP38:CP90 into DE38:DE90

Group 3: HOB=3: CP91:CD330 into DF91:DF330 etc

This is a manual process – the lab instructions use a formula approach instead

	A	DD	DE	DF	DG	DH
1	Variable Group	Enviro Domain	Enviro Domain	Enviro Domain	Enviro Domain	Enviro Domain
2	Variable	HOB=1	HOB=2	HOB=3	HOB=4	HOB=5
3	Question					
4	Variable coding:	Enviro Domain HOB=1	Enviro Domain HOB=2	Enviro Domain HOB=3	Enviro Domain HOB=4	Enviro Domain HOB=5

NB you are using Employee Domain in the Lab!

34	P943	3.75				
35	P943	3.75				
36	P946	3.5				
37	P982	2				
38	P14		4			
39	P167		3.5			
40	P291		3			
41	P356		4			
42	P439		4			
89	P1016		4.25			
92	P54			3.5		
93	P70			5		
94	P71			4.75		
588	P1030				3.25	
589	P1034				4.5	
590	P1038				3	
591	P246				4.25	
592	P24					3.75
593	P84					4.25
594	P92					5
595	P94					4.75
596	P95					4.25

68 2) In SustainabilityData cell CX5, enter: **=IF(\$AY5=1,\$CT5,"x")**. Copy this to cell CY5, and change =1 to =2

Input Range:

\$DD\$5:\$DH\$745

Grouped By:

☒ Columns

☐ Rows

☐ Labels in first row

Alpha:

0.05

Lab 8 Task 3

	A	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW	CX	CY	CZ	DA	DB	DC	DD	DE	DF	DG	DH
1	Variable Gender		Societal Domain	Environmental Domain	Supplier Domain	Shareholder Domain	Community Domain	Employee Domain	Customer Domain	Sustainability literacy		Employee Domain HOB=1	Employee Domain HOB=2	Employee Domain HOB=3	Employee Domain HOB=4	Employee Domain HOB=5		Enviro Domain HOB=1	Enviro Domain HOB=2	Enviro Domain HOB=3	Enviro Domain HOB=4	Enviro Domain HOB=5
2	Variable Gender		Societal Domain	Environmental Domain	Supplier Domain	Shareholder Domain	Community Domain	Employee Domain	Customer Domain	Sustainability literacy												
3	Question	What gender do you identify with?	1: Not at all important 5: Extremely Important	1: Not at all important 5: Extremely Important	1: Not at all important 5: Extremely Important	1: Not at all important 5: Extremely Important	1: Not at all important 5: Extremely Important	1: Not at all important 5: Extremely Important	1: Not at all important 5: Extremely Important	1: Strongly disagree 3: Neither disagree nor agree 5: Strongly agree												
4	Variable coding:																					
5	P97	1	3	4	4	4.333333	3.5	4.333333	4	2.222222												
6	P157	1	3	1.5	3.666667	4.666667	3.25	4.333333	1.25	2.111111												
7	P171	1	3.5	1.5	3.333333	3	3.5	4	2.75	1.555556												
8	P198	1	1	1	2.333333	3.666667	2.5	5	1	1												
9	P275	1	4	1	1	1	5	5	4.75	2.444444												
10	P276	1	1	1.5	4.333333	4.666667	3.5	4	1	1.777778												
11	P344	1	3	3	3	3	3	3	3	3												
12	P358	1	2.25	5	4.666667	4	3.75	5	3.25	2.333333												
13	P391	1	1	5	1	1	2	3.666667	3	3												
14	P410	1	3	3	3	3	3	3	3.25	3												
15	P448	1	2.25	3.5	4.333333	4.666667	3.25	4.333333	3.25	1.444444												
16	P541	1	3	1	3	3	3	3	1	1.666667												
17	P579	1	3	4	4	4	3.5	4.333333	3.25	3.333333												
18	P620	1	3.5	5	3.666667	3.666667	3.75	4.333333	3.5	2.444444												
19	P652	1	3	3	3	3	3	3	3	2.333333												
20	P690	1	4	4.75	4.333333	5	4.25	5	4	4.111111												
21	P767	1	3	3	3	3	3	3	3	2.666667												
22	P819	1	1	2.75	1.666667	3.666667	1	1	1.5	2.333333												
23	P938	1	2.25	3.75	4.666667	3.666667	2.75	5	3.25	2.444444												
24	P947	1	1	3.75	2.333333	3.333333	2.5	3.333333	3	1												
25	P91	2	3.25	4.75	4.333333	4	4	4.333333	4.25	2.777778												
26	P174	2	2.75	3.5	3	4.333333	2.75	3.666667	3.75	2.777778												
27	P183	2	3	3.5	3	3	3	3	3	3.222222												
28	P405	2	3.25	4	3	3	4.25	5	4.5	2.555556												
29	P470	2	3	5	5	5	4	5	4	3.222222												

Formula approach used in the lab
(but you might need to use the manual method
if the formula approach does not work)

See video in Stream:
Lab 8 Task 3
<https://webcast.massey.ac.nz/Mediasite/Plaly/fdc1201ede7345659b6306e3f11cea271d>

Formula approach used in the lab
(but you might need to use the manual method
if the formula approach does not work)

See video in Stream:
Lab 8 Task 3

<https://webcast.massey.ac.nz/Mediasite/Play/fdc1201ede7345659b6306e3f11cea271d>

ANOVA in Excel:

separate into Subsamples

Anova: Single Factor		Eg for Environmental Domain (Lab: Employee Domain)			Enviro Domain HOB=1	Employee Domain
SUMMARY						
Groups	Count	Sum	Average	Variance		
Column 1	33	109.25	3.31060606	1.68371212		
Column 2	53	221.5	4.17924528	0.4792725		
Column 3	240	992.75	4.13645833	0.45541209		
Column 4	261	1104.75	4.23275862	0.37758621		
Column 5	154	679.25	4.41071429	0.39107726		
ANOVA						
Source of Variati	SS	df	MS	F	P-value	F crit
Between Gro	34.1835064	4	8.54587661	18.1968308	2.8906E-14	2.38403199
Within Group	345.651682	736	0.46963544			
Total	379.835189	740				

NB: 2.8906E-14 => 13 0s before 2
=0.0000000000000028906

$$F_c = 2.384 \quad F = 18.2 \text{ (rounded)}$$

∴ significant: at least one significant difference between the five levels of HowOftenBuy and Environmental Domain

But which? → t-test of pairs of means (1 vs 2, 2 vs 3 etc)

Z-test of Difference: Proportions

Probability of being a voter in an election, among a general population sample of 188: 0.425

Probability of being a voter, among those that regularly read newspapers: 0.479

1) Establish null and alternative hypotheses

H_0 : people are not more likely to have voted in the election if they read a newspaper;

H_a : people are more likely to have voted in the last general election if they read a newspaper

2) Decide critical level of significance

$$\alpha = 0.05 \text{ (ie, 5\% in right tail)} \therefore z_c = 1.645$$

3) Calculate Z:
$$\frac{p - P}{\sqrt{\frac{P(1-P)}{n}}} = \frac{0.479 - 0.425}{\sqrt{\frac{0.425(1-0.425)}{188}}} = \frac{0.054}{\sqrt{\frac{0.244}{188}}} = \frac{0.054}{0.036} = 1.50$$

4) Compare against critical level of significance: $1.50 < 1.645$

5) Cannot reject H_0

Multivariate: Proportions

Probability people aged 22-25 support a tax decrease: 0.508 (n=61)

Probability people aged 30-39 support a tax decrease: 0.613 (n=142)

1) H_0 : Support for tax decrease does not vary between the two age groups (22-25, 30-39)

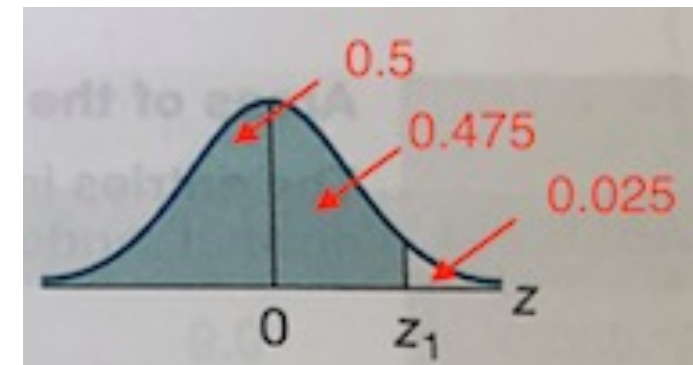
H_A : People aged 30-39 are more likely to support tax decrease

2) Decide critical level of significance

$$\alpha = 0.05 \text{ (ie, 5\% in right tail)} \therefore z_c = 1.645$$

3) Calculate Z

$$Z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\hat{p}(1 - \hat{p}) \left[\frac{n_1 + n_2}{n_1 n_2} \right]}}$$



$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

Multivariate: Proportions

$$\hat{p} = \frac{142(0.613) + 61(0.508)}{142 + 61} = 0.581$$

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$Z = \frac{(0.613 - 0.508) - (0)}{\sqrt{0.581(1 - 0.581) \left[\frac{142 + 61}{(142)(61)} \right]}} = 1.4 \quad Z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\hat{p}(1 - \hat{p}) \left[\frac{n_1 + n_2}{n_1 n_2} \right]}}$$

4) Compare against critical level of significance: $1.4 < 1.645$

5) Cannot reject null hypothesis, i.e. cannot say support for tax decrease varies by age group

Filter Grouping vs Recoding: Nested IF statements

We have grouped Income
and Age into fewer groups
within PivotTables
BUT these are temporary
(stored in cache, not the file)

To make them permanent we need to **recode**
Which requires using nested IF statements

Copy Column CM, insert copied cells into new CW
Rewrite CW such that:
=IF(CM5=1,1, IF(CM5=2,1,
IF(CM5=3,2,IF(CM5=4,2, IF(CM5=5,3,IF(CM5=6,3,""))))))
and copy down to CW1046

Then Sort, based on the new column CW
(Make sure the whole data range (eg A5:CW1046) is selected, then sort Smallest to Largest

CM	CW
What is your total household income?	Household income literacy
1: Up to \$30,000	1: Lower
2: \$30,001 - \$50,000	2: Middle
3: \$50,001 - \$70,000	3: Upper
4: \$70,001 - \$100,000	
5: \$100,001 - \$130,000	
6: Over \$130,000	
7: Would rather not say	
Household Income	Income Recoded
4	2
6	3
6	3
4	2
5	3
5	3
6	3
3	2
2	1
5	3
5	3
1	1
1	1

Lab 8

115.109 Topic 8 Lab Worksheet: Comparing Means of Composite Variables

Due date: October 5

Version: 1 (3.09)

Rationale:
This lab con
and 7), but i
from August
will need to

Task 1: Generate descriptive statistics for the composite variables used in Lab 7, for SustainabilityData by Literacy

There is a fifth method (beyond the four listed in Task 2 of Lab 7) that can be used to generate subsample descriptive statistics.

In previous labs we have used =COUNTIF(range, criteria), where criteria could be Sustainability knowledge = "5" [see Lab 2]; and =SumIf (Lab 4 SKUs)

And =AVERAGEIF (range, criteria, [average_range]), where average_range is what you want to take the average of

Excel trainer 'Exceljet' (Dave and Lisa of Colorado) explain it here: <https://exceljet.net/excel-functions/excel-averageif-function>

There is no =STDEVIF() function, but you can use a nested IF statement as =STDEV(IF(... instead: =STDEV(IF(range=criteria, target_range), and the same for =average(IF(and =count(IF(

- 1) We will use the composite variable 'Literacy' to analyse the others, dividing into "Higher Literacy" (above mean, calculated in cell CV1068) and "Lower Literacy" levels (Use these three formulae to create descriptive statistics in the SustainabilityData sheet. I've entered the formulas for Higher:Societal; complete the rest.
- 2) In the SustainabilityData sheet; cells CO1069:CO1071: copy and paste the descriptive stats for each composite variable for the (above mean) Higher Literacy group
This uses the formula =AVERAGE(IF(SustainabilityData!\$CV\$5:\$CV\$1046>\$CV\$1068,SustainabilityData!CO\$5:CO\$1046)) for Mean for Societal for the Higher literacy group
For StDev Societal:Higher the formula is =STDEV(IF(SustainabilityData!\$CV\$5:\$CV\$1046>\$CV\$1068,SustainabilityData!CO\$5:CO\$1046))
For sample size Societal:Higher: =COUNT(IF(SustainabilityData!\$CV\$5:\$CV\$1046>\$CV\$1068,SustainabilityData!CO\$5:CO\$1046)) [Count result should be the same across the domain]
- 3) Copy the formulas in CO1069:CO1071, and paste into CO1072 (this will also paste into CO1070 and CO1071), and update for below the mean: change the > to < for each formula, then copy across
- 4) Once complete in the SustainabilityData sheet, copy CO1069:CV1071, and Paste Special > Values and Transpose in cell D20 below. Repeat for CO1072:CV1074, and Paste Special > Values and Transpo:

Column	Composite Variable	Higher literacy			Lower literacy		
		Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size
CO	Societal Domain						
CP	Environmental Domain						
CQ	Supplier Domain						
CR	Shareholder Domain						
CS	Community Domain						
CT	Employee Domain						

- 5) Now copy down the Z-test calculations in D31:D38, the t-test calculations in E31:H38, and complete the interpretation box

Column	Composite Variable	Z-test statistic	t-test numerator	Pooled stdev	t-test denominator	t-test statistic	critical value of t
CO	Societal Domain	#DIV/0!	0.000	0.000	#DIV/0!	#DIV/0!	#NUM!
CP	Environmental Domain						#NUM!
CQ	Supplier Domain						#NUM!
CR	Shareholder Domain						#NUM!
CS	Community Domain						#NUM!
CT	Employee Domain						#NUM!
CU	Customer Domain						#NUM!
CV	Sustainability literacy						#NUM!

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Lab 8

5) Now copy down the Z-test calculations in D31:D38, the t-test calculations in E31:H38, and complete the interpretation box

Column	Composite Variable	Z-test statistic	t-test numerator	Pooled stdev	t-test denominator	t-test statistic	critical value of t
CO	Societal Domain	#DIV/0!	0.000	0.000	#DIV/0!	#DIV/0!	#NUM!
CP	Environmental Domain						#NUM!
CQ	Supplier Domain						#NUM!
CR	Shareholder Domain						#NUM!
CS	Community Domain						#NUM!
CT	Employee Domain						#NUM!
CU	Customer Domain						#NUM!
CV	Sustainability literacy						#NUM!

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Task 2: Contrast the SustainabilityData descriptive statistics with those for ClassData

This is far simpler to generate the base data, as we are not doing any subsample analyses here. However, you do need to repeat the creation of the composite variables in cells C05:CV5, and copy down to row 372 in the ClassData sheet.

- Copy and Paste Special > Formulas cells D48:D55 (this sheet) into E48 and while E48:E55 is selected, click Home ribbon >> Find & Select >> Replace>> Find: average; Replace with: stdev; click Replace All [8 replacements]. Repeat into F48:F55, this time replacing average with count
- Copy and Paste Special > Formulas cells D48:F55 into G48 and while G48:I55 is selected, Find and replace Sustainability with Class and Replace All [24 replacements];
- Without closing the dialog box (or click Find&Select again), Find and Replace 1046 with 372 [24 replacements].
- Now complete the z- and t-tests boxes in N49:O56, and the interpretation box: note which are significant, and compare the Z and t-test results.

[NOTE: I have cleaned the data to remove any responses that did not complete the Domain questions. If there were any blank rows, these calculations would fail.]

Column	Composite Variable	Variable Mean	Standard Deviation	Variable Sample Size	Variable Mean	Standard Deviation	Variable Sample Size
CO	Societal Domain	#DIV/0!					
CP	Environmental Domain	#DIV/0!					
CQ	Supplier Domain	#DIV/0!					
CR	Shareholder Domain	#DIV/0!					
CS	Community Domain	#DIV/0!					
CT	Employee Domain	#DIV/0!					
CU	Customer Domain	#DIV/0!					
CV	Sustainability literacy	#DIV/0!					

Find: average

Within: Sheet

Search: By Rows

Replace with: stdev

Replace

Replace All

Find what: SustainabilityData

Replace with: ClassData

Replace

Replace All

Z-test statistic	t-test statistic
#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!
	#DIV/0!
1.960	#NUM!

Critical values of z and t

<<Z- and t-test results interpretation>>

Excel Options

General

Formulas

Proofing

Save

Language

Advanced

Customize Ribbon

Quick Access Toolbar

Add-on

Trust Center

(Windows)

Lab 8

Task 3: Use the Excel Data Analysis Toolpak (ANOVA) to test the Employee Domain composite by How Often Buy (SustainabilityData, column AY)

If activated as per Lab 7, you will see a Data Analysis group at the right of the Data tab ribbon.

[Activating Mac: Tools menu > Excel Add-Ins > Analysis ToolPak]

[Activating Windows: File menu > Options > Add-ins > Excel Add-ins > Go > Analysis ToolPak]

ANOVA requires the data to be tested to be in separate columns. Preparing the data is tedious if you do it manually (by sorting, duplicating, then deleting), but there is a simpler way:

- 1) Check that your data is still sorted by HowOftenBuy (column AY), smallest to largest, rows 5:1046 (same as for Lab 7 Task 4 - if not, sort it again)
- 2) In SustainabilityData cell CX5, enter: `=IF($A$5=1,$C$5,"x")`. Copy this to cell CY5, and change =1 to =2
Copy to cell CZ5, and change to =3; copy to DA5, and change to =4; copy to DB5 and change to =5. Then copy CX5:DB5 and paste from CX6:CX1046
- 3) ANOVA cannot cope with non-numeric characters, so we need to remove the "x" values. Copy CX5:DB1046
and while the whole range is selected, Paste Special > Values, replacing the formulas with their results.
Now Find & Select > Replace, and put x into the Find box; leave the Replace box blank; click on Replace All. There should be 4469 replacements
- 4) You should end up with a staggered set of columns, where we have broken column CT into five non-overlapping columns.
- 4) In cell CX1050 use `=COUNT(CX5:CX1046)` and copy across to DB1050. If you (and I) have done this correctly, your counts should be: 33, 53, 240, 261, 154
- 5) Now call up the Data >>> Analysis Toolpak >>> ANOVA: Single Factor, and complete as Input Range: `$CX$5:$DB$1046`
The results will appear in a new sheet. Copy and Paste Special > Values into the Summary and ANOVA tables below, and interpret the ANOVA result in the box.

Anova: Single Factor

SUMMARY

Groups Count Sum

Column 1 Column 2 Column 3 Column 4 Column 5

This is a formula-based process which *should* be easier than a manual copy-and-paste process
(but that is a backstop if this doesn't work for you – see video in Topic 8 if needed)

Column 1 Column 2 Column 3 Column 4 Column 5

ANOVA

Source of Variat	SS	df	MS	F	P-value

Excel Options

General
Formulas
Proofing
Save
Language
Advanced
Customize Ribbon
Quick Access Toolbar
Add-ins

View and manage Microsoft Office Add-ins.

Name	Location	Type
Active Application Add-ins		
Acrobat PDFMaker Office COM Addin	C:\...C\PDFMaker\Office\PDFMOfficeAddin.dll	COM Add-in
PhantomPDF Creator COM Addin	C:\...gins\Creator\vb6\FPC_ExcelAddin_x86.dll	COM Add-in
Template for Excel	C:\...ple for Office\Temple.Excel.vstolystolocal	COM Add-in
Inactive Application Add-ins		
Analysis ToolPak	C:\...Office16\Library\Analysis\ANALYSIS32.XLL	Excel Add-in
Analysis ToolPak - VBA	C:\...ffice16\Library\Analysis\ATPVBAEN.XLAM	Excel Add-in
Euro Currency Tools	C:\...Office\Office16\Library\EUROTOOL.XLAM	Excel Add-in
Inquire	C:\...soft Office\Office16\DCF\NativeShim.dll	COM Add-in
Microsoft Actions Pane 3		XML Expansion Pack
Microsoft Power Map for Excel	C:\...ap Excel Add-in\EXCELPLUGINSHELL.DLL	COM Add-in
Microsoft Power Pivot for Excel	C:\...el Add-in\PowerPivotExcelClientAddin.dll	COM Add-in
Microsoft Power View for Excel	C:\...el Add-in\AdHocReportingExcelClient.dll	COM Add-in
Solver Add-in	C:\...Office16\Library\SOLVER\SOLVER.XLAM	Excel Add-in

Document Related Add-ins

No Document Related Add-ins

Add-in: Acrobat PDFMaker Office COM Addin
Publisher: Adobe Inc.
Compatibility: No compatibility information available
Location: C:\Program Files (x86)\Adobe\Acrobat

Description: Acrobat PDFMaker Office COM Addin

Manager: Excel Add-ins

Go...

Add-ins available:

☒ Analysis ToolPak
☐ Analysis ToolPak - VBA
☐ Euro Currency Tools
☐ Solver Add-in

OK

Cancel

Task 4: Complete confidence intervals around the five means above.

What can you interpret about the difference in means, from these confidence intervals?

Lab 8

Task 5: Use the Analysis Toolpak option t-Test Two-Sample (Assuming Equal Variances), testing the Employee Domain composite by How Often Buy =4 vs =5

- 1) Use column DA as the Variable 1 range and DB as the Variable 2 range
Use $\alpha = 0.05$ (1 chance in 20 of making a Type 1 error: incorrectly rejecting a true Null Hypothesis)
Copy results below, and interpret

Copy results here: t-Test: Two-Sample Assuming Equal Variances

	Variable 1	Variable 2	Employee Domain
Mean			
Variance			
Observations			
Pooled Variance			
Hypothesized Mean Difference			
df			
t Stat			
P(T<=t) one-tail			
t Critical one-tail			
P(T<=t) two-tail			
t Critical two-tail			

t-Test: Two-Sample Assuming Equal Variances

Input

Variable 1 Range: \$DA\$5:\$DA\$1046

Variable 2 Range: \$DB\$5:\$DB\$1046

Hypothesized Mean Difference: 0

☐ Labels

Alpha: 0.05

Output options

☐ Output Range:

☒ New Worksheet Ply:

New Workbook

NOTE!! You need to activate (run) each tool while viewing the datafile (PerceptiveData), not this sheet!!

Enter 0 (null hypothesis: no difference in means)

Default value - you can change it, but don't!

Data Analysis

Analysis Tools

- Random Number Generation
- Rank and Percentile
- Regression
- Sampling
- t-Test: Paired Two Sample for Means
- t-Test: Two-Sample Assuming Equal Variances**
- t-Test: Two-Sample Assuming Unequal Variances
- z-Test: Two Sample for Means

Lab 8

Task 6: Create PivotChart

- 1) Select cells SustainabilityData A4:CV1046 >> Insert >> PivotChart
- 2) Drag the Societal Domain variable into the Row Labels
- 3) Drag How Often Buy into the Column Labels
- 4) Change Sum of Societal to Average
- 5) Modify the chart from bar to line
- 6) Add in the remaining Domain variables
- 7) How Often Buy includes the variable Household Income (Don't use the Filter in the PivotChart)
- 8) Modify the Y axis to better represent the data (ideally you would indicate the units)
- 9) Copy chart below

Bonus Task 7: replicate the Task 6 chart

The screenshot shows the Excel interface with the 'Create PivotChart' and 'PivotTable Field' tasks. The 'Create PivotChart' dialog box is open, showing the 'Table/Range' as 'Perceptived: SustainabilityData!\$A\$4:\$CV\$1046'. The 'PivotTable Field' task is also open, showing the 'Source field: Societal Domain' and the 'Field name: Average of Societal Domain'. The 'Summarise by' dropdown is set to 'Average'. The 'PivotChart Fields' task is open, showing the 'HowOftenBuy' variable in the 'Column Labels' and the 'Societal Domain' variables in the 'Row Labels'. The chart is a line chart. The 'PivotChart Fields' task is also open, showing the 'HowOftenBuy' variable in the 'Column Labels' and the 'Societal Domain' variables in the 'Row Labels'. The chart is a line chart. The 'PivotChart Fields' task is also open, showing the 'HowOftenBuy' variable in the 'Column Labels' and the 'Societal Domain' variables in the 'Row Labels'. The chart is a line chart.